

Supplementary material for Eisenstein criterion

Theorem (Eisenstein) Let $a_0, a_1, \dots, a_n$ be integers. **Eisenstein's Criterion** states that the polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

cannot be factored into the product of two non-constant polynomials if:

1. p is a prime that divides each of $a_0, a_1, \dots, a_{n-1}$;
2. a_n is not divisible by p ;
3. a_0 is not divisible by p^2 .

Problem (Optional) Prove Eisenstein's Criterion.

Problem (Optional) Show that there are irreducible polynomials which Eisenstein's criterion cannot detect. More precisely, find an irreducible polynomial $f(x)$ for which Eisenstein's criterion fails for all primes p , and for all shifts $f(x - a)$.

Example (A very important example you may encounter in future) We shall use Eisenstein's Criterion to prove that $f(x, y, z) = x^2 + y^2 + z^2$ is irreducible over $\mathbb{C}[x, y, z]$.

Consider the (UFD) ring $\mathbb{A} = \mathbb{C}[y, z]$ and the expression

$$x^2 + (y + iz)(y - iz) \in \mathbb{A}[x].$$

Since:

1. $(y + iz)$ is prime in $\mathbb{A}$ and divides $a_1 = 0$;
2. $a_n = 1$ is not divisible by $(y + iz)$;
3. $a_0 = y^2 + z^2$ is not divisible by $(y + iz)^2$,

we conclude that f is irreducible over $\mathbb{C}[x, y, z]$.

Theorem (Eisenstein, extended) Let $a_0, a_1, \dots, a_n$ be integers. **Extended Eisenstein's Criterion** states that

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

has an irreducible factor with $\deg > k$ if:

1. p is a prime that divides each of $a_0, a_1, \dots, a_k$;
2. a_{k+1} is not divisible by p ;
3. a_0 is not divisible by p^2 .

Problem (Optional) Prove Extended Eisenstein's Criterion.

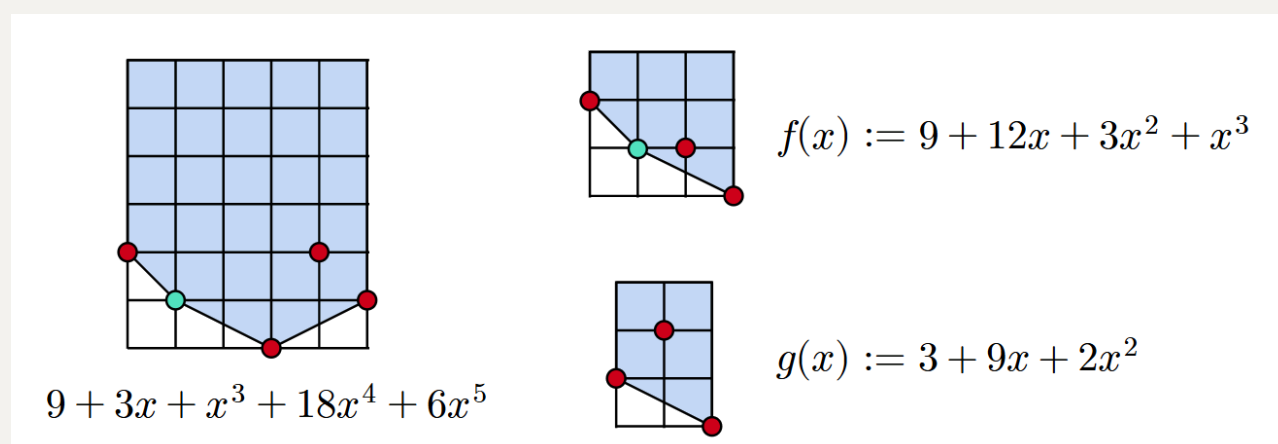
Example (IMO 1993) Given an integer $n \geq 1$, consider the polynomial $f(x) = x^n + 5x^{n-1} + 3$. Show that f is irreducible in $\mathbb{Z}[x]$.

Taking $p = 3$ and $k = n - 2$ in Extended Eisenstein's Criterion, we see that f has some irreducible factor g with $\deg g > (n - 2)$.

We are done when $\deg g = n$.

When $\deg g = n - 1$, there exists some $d \in \mathbb{Z}$ such that $(x - d) \cdot g(x) = f(x)$. We learn from middle school that $d \in \{\pm 1, \pm 3\}$. As a result, $f(d)$ is an odd number (absurd).

Definition I call it mod 3-convex hull of polynomials, guess what it means?



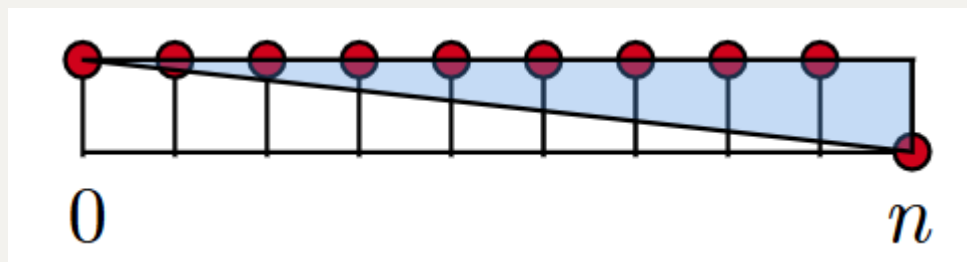
Problem (optional) Create the corresponding mod 3-convex hull for the product polynomial $f(x) \cdot g(x)$, and count the number of integer points where the **bottom polygonal chain** passes through.

Theorem (Eisenstein, diagrammatic) Let f be a polynomial with $f(0) \neq 0$. If there exists some prime p , such that

- the bottom polygonal chain is straight and passes through none of the integral points,

then the polynomial is irreducible in $\mathbb{Q}[x]$.

Problem (optional) Prove Eisenstein Criterion **without words**.



Example (某年南开 (?) 考研压轴题) Create a criterion (Eisenstein-analogue) according to the following diagram, and make the problem difficult:

